

6.5 Medical imaging

This section provides knowledge and understanding of X-rays, CAT scans, PET scans and ultrasound scans. This section shows how the developments in medical imaging have led to a number of valuable non-invasive techniques used in hospitals.

Not all hospitals in this country are equipped with complex scanners. Learners have the chance to discuss the ethical issues in the treatment of humans and the ways in which society uses science to inform decision making (HSW10 and 12).

6.5.1 Using X-rays

Learning outcomes	Additional guidance
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Learners should be able to demonstrate and apply their knowledge and understanding of:

- (a) basic structure of an X-ray tube;
components – heater (cathode), anode, target metal and high voltage supply

- (b)** production of X-ray photons from an X-ray tube
- (c)** X-ray attenuation mechanisms; simple scatter, photoelectric effect, Compton effect and pair production
- (d)** attenuation of X-rays; $I = I_0 e^{-\mu x}$, where μ is the attenuation (absorption) coefficient *M0.5, M3.11*
- (e)** X-ray imaging with contrast media; barium and iodine HSW9, 10, 12
- (f)** computerised axial tomography (CAT) scanning; components – rotating X-tube producing a thin fan-shaped X-ray beam, ring of detectors, computer software and display
- (g)** advantages of a CAT scan over an X-ray image. HSW9, 10, 12